

Forecasting Australian Real GDP Growth

Group 17

Question and main result

Research question

Which model best predicts year-on-year real GDP growth in Australia over the last 10 years of the sample: an export-based ADL, an ADL with two GDP lags, a univariate AR(2), a small VAR(2), or an equal-weight forecast combination?

Forecasting setup

- Quarterly Australian macroeconomic data.
- Recursive 1-step-ahead forecasts.
- Clean common-support evaluation window over the last 40 quarters.

Takeaway: The benchmark combination is slightly best, but the ToT-based ADL is the strongest single-equation forecast once alternative regressors are considered.

Main result

- In the benchmark horse race, the **combination is marginally best** on MSFE and RMSE.
- Among ADL models, the **ToT-based ADL performs best** out of sample.
- ADL(2) is dominated by the simpler ADL specifications.

Economic message

GDP growth is persistent. The external regressor matters: terms of trade outperform exports out of sample, while the short rate remains weak.

Data, variables, and transformations

Variable	Effective series used	Transformation	Role
Real GDP	AUSGDPRQDSMEI	$100 \times (\text{GDP}_t / \text{GDP}_{t-4} - 1)$	Dependent variable
Consumer prices 3-month rate	AUSCPIALLQINMEI IR3TIB01AUQ156N	$100 \times (\text{CPI}_t / \text{CPI}_{t-4} - 1)$ Level in percent	Inflation Monetary indicator
Real exports	NXRSAXDCAUQ	$100 \times (\text{EXP}_t / \text{EXP}_{t-4} - 1)$	Baseline external regressor
Terms of trade	AUSLOCOTTNOSTSAM	$100 \times (\text{ToT}_t / \text{ToT}_{t-4} - 1)$	Robustness external regressor

Sample

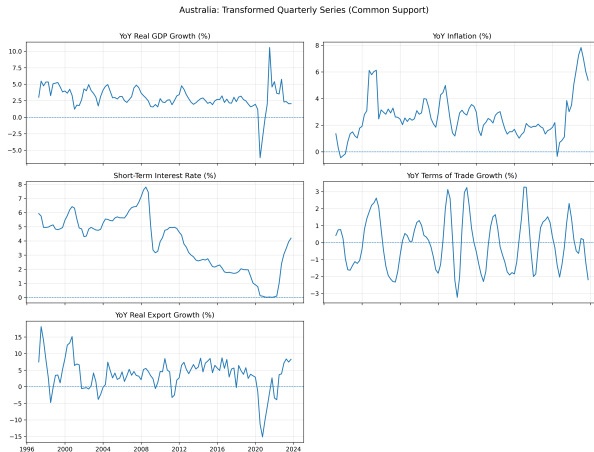
- Cleaned quarterly sample with exact common support.
- Estimation sample for the preferred ADL: 105 observations.
- Forecast evaluation based on the last 40 quarters.

Cleaning choices

- Exact common support across all quarterly series.
- Year-on-year growth rates built with `pct_change(fill_method=None)`.
- No artificial endpoint padding in GDP growth.

The final analysis keeps exports as the baseline external regressor, but also tests terms of trade and the change in the short rate as disciplined robustness alternatives.

Descriptive evidence



GDP growth is stable for much of the sample, then collapses and rebounds during the pandemic. Inflation and short rates rise sharply after 2021. The external sector is materially more volatile than domestic inflation.

Descriptive statistics and stationarity

Main descriptive evidence

Variable	Main feature	ADF reading
GDP growth	Persistent	Weak rejection
Inflation	Moderate volatility	Borderline rejection
Interest rate	Level variable	Least comfortable
Export growth	More volatile	Stationary

Interpretation

- GDP growth remains persistent even after year-on-year transformation.
- Inflation is closer to stationarity than the interest rate in levels.
- Export growth is stationary and highly volatile.
- Terms of trade are explored as an alternative external indicator in the robustness analysis.

Takeaway: The cleaned sample is more defensible econometrically, but the interest-rate level and the persistence of GDP growth still require caution in interpretation.

Model class and preferred ADL specification

Competing models

$$\text{ADL(1): } g_t = \alpha + \phi g_{t-1} + \beta x_{t-1} + \gamma i_{t-1} + \delta \pi_{t-1} + u_t$$

$$\text{ADL(2): } g_t = \alpha + \phi_1 g_{t-1} + \phi_2 g_{t-2} + \beta x_{t-1} + \gamma i_{t-1} + \delta \pi_{t-1} + u_t$$

$$\text{AR(2): } g_t = \alpha + \rho_1 g_{t-1} + \rho_2 g_{t-2} + e_t$$

$$\text{VAR(2): } y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t,$$

with $y_t = (g_t, \pi_t, i_t)'$ and x_t denoting the external regressor.

Within-family selection

- baseline ADL: lagged export growth, lagged inflation, lagged interest;
- ADL(2): same regressors plus g_{t-2} ;
- robustness ADLs: replace exports with ToT, or replace the rate level with Δi_{t-1} .

Takeaway: The baseline export ADL is preferred in sample, but alternative regressors are tested because forecasting performance may differ out of sample.

In-sample fit comparison

Model	AIC	BIC	Adj. R^2
Contemp. ADL	370.8	384.1	0.365
Export ADL	369.7	382.9	0.372
ToT ADL	372.2	385.5	0.357
Δi ADL	371.0	384.3	0.364
ADL(2)	371.7	387.6	0.366

The export-based ADL remains the best *in-sample* specification, while ADL(2) and the alternative regressors are kept as disciplined robustness checks.

Preferred ADL estimates

HAC-robust coefficient estimates: export ADL

Regressor	Coef.	Std. err.	<i>p</i> -value
Intercept	1.2665	0.657	0.057
g_{t-1}	0.6231	0.089	0.000
x_{t-1}	-0.0529	0.021	0.013
i_{t-1}	0.0974	0.071	0.174
π_{t-1}	-0.1336	0.063	0.037

Interpretation

- **Persistence dominates.** Lagged GDP growth is the strongest regressor.
- **Inflation matters.** The lagged inflation coefficient is negative and statistically significant.
- **Export growth is informative.** It contains reduced-form predictive content, though not a structural causal interpretation.
- **The short rate remains weak.**

Takeaway: The export-based ADL is still mainly a persistence model, but the robustness analysis will show that terms of trade forecast slightly better out of sample.

Diagnostics and robustness

Preferred ADL diagnostics

Test	Reading
Breusch-Godfrey LM(4)	residual dependence remains
White LM	some heteroskedasticity remains
Jarque-Bera	strong non-normality

Crisis-dummy robustness

Pandemic pulse dummies improve fit and residual behavior substantially:

$$\text{AIC} = 323.3, \quad \text{Adj. } R^2 = 0.603.$$

Takeaway: The short-rate effect remains statistically weak, while crisis dummies materially improve the model's in-sample credibility.

Wald test on the short rate

- Null hypothesis: $i_{t-1} = 0$
- Wald statistic: 1.715
- p -value: 0.193
- Interpretation: the short-rate term has no clear marginal predictive effect in the preferred ADL

Conclusion

The main residual problem is crisis-period non-normality rather than generic heteroskedasticity. Crisis dummies improve the model without changing the main forecasting ranking.

Forecast design

Pseudo-out-of-sample protocol

- Expanding recursive window.
- 1-step-ahead point forecasts.
- Models re-estimated at each forecast origin.
- Forecast comparison over the last 40 quarters.

Models compared

- benchmark ADL(1), ADL(2), AR(2), VAR(2), and the equal-weight combination;
- robustness ADLs with terms of trade and with Δi_{t-1} .

Combination rule

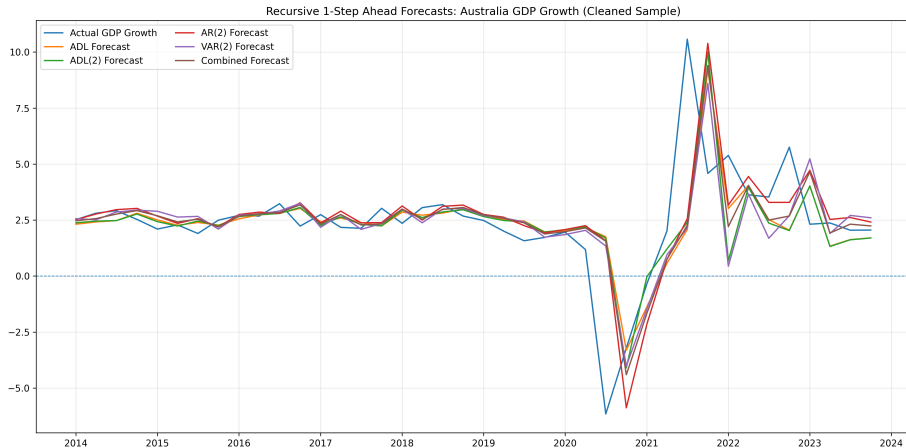
$$\hat{g}_t^{comb} = \frac{1}{3} \left(\hat{g}_t^{ADL} + \hat{g}_t^{AR(2)} + \hat{g}_t^{VAR(2)} \right).$$

Why test ADL(2)?

Quarterly GDP growth can exhibit residual persistence beyond one lag. The second GDP lag is therefore tested directly, but it is retained only if it improves out-of-sample forecasting performance.

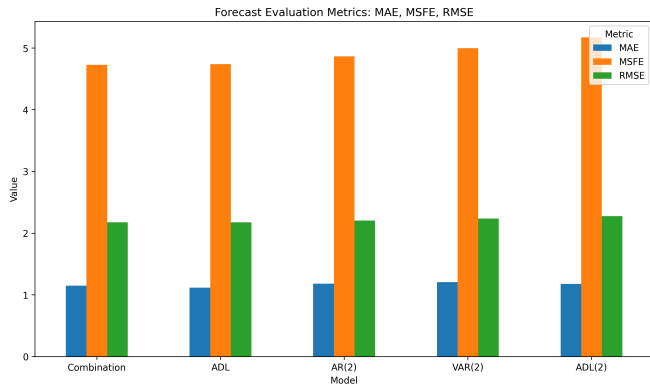
This design uses exact common support only, so the old endpoint artifact is gone.

Recursive 1-step-ahead forecasts



The figure overlays actual GDP growth with the benchmark models and the two alternative-regressor ADLs. The ToT-based ADL tracks the realized path slightly better than the export-based specification, while ADL(2) does not improve enough to justify the extra lag.

Forecast accuracy comparison



Benchmark horse race

Model	MAE	MSFE	RMSE
Combination	1.148	4.727	2.174
ADL	1.118	4.737	2.176
AR(2)	1.180	4.861	2.205
VAR(2)	1.207	4.994	2.235
ADL(2)	1.177	5.171	2.274

Ranking

Combination < ADL < AR(2) < VAR(2) < ADL(2)

The explicit lag-choice comparison rejects ADL(2): the second GDP lag does not improve out-of-sample performance.

Takeaway: Within the required benchmark horse race, the combination is marginally first; the export-based ADL remains the best individual benchmark and clearly beats ADL(2).

Diebold-Mariano tests of predictive accuracy

Null hypothesis

The Diebold-Mariano test evaluates whether two competing forecasts have equal expected squared forecast loss.

Pairwise benchmark results

Comparison	DM stat.	<i>p</i> -value	Conclusion
ADL(1) vs AR(2)	-0.264	0.793	No significant difference
ADL(1) vs VAR(2)	-0.463	0.646	No significant difference
ADL(1) vs Comb.	0.046	0.963	No significant difference
ADL(1) vs ADL(2)	-0.902	0.373	No significant difference

Takeaway: Among the required benchmark models, predictive-accuracy differences are economically small and statistically inconclusive.

Main implication

- None of the benchmark forecast-accuracy gaps is statistically significant.
- Hence, the benchmark RMSE ranking should not be interpreted as decisive.
- In particular, the combination is not significantly more accurate than the export-based ADL.

This supports using a parsimonious ADL benchmark rather than chasing negligible differences in benchmark RMSE.

Why retain the ADL as the preferred single-model class?

Why ADL remains the benchmark class

- ➊ **Parsimony.** ADLs remain much smaller than the VAR and avoid unnecessary estimation noise.
- ➋ **Persistence.** Lagged GDP growth captures the dominant forecasting signal in the data.
- ➌ **Interpretability.** External indicators and inflation provide a clear reduced-form narrative.

Among the ADL variants, the **choice of external regressor matters**: terms of trade forecast better than exports out of sample, even though exports fit slightly better in sample.

Takeaway: The ADL remains the preferred single-model class, and within that class the ToT specification is the strongest out-of-sample performer.

Alternative-regressor comparison

ADL variant	MAE	MSFE	RMSE
ToT	1.085	4.496	2.120
Δi	1.121	4.587	2.142
Exports	1.118	4.737	2.176

DM tests do not reject equal predictive accuracy relative to the export-based ADL:

$$p_{\text{ToT}} = 0.560, \quad p_{\Delta i} = 0.684.$$

So ToT is numerically best out of sample, but not statistically decisive.

Conclusion

- ❶ **Benchmark result:** in the required horse race, the equal-weight combination delivers the lowest RMSE and MSFE, but only by a very small margin over the export-based ADL.
- ❷ **Robustness result:** once alternative regressors are considered, the terms-of-trade ADL achieves the lowest forecast errors among the ADL variants and numerically outperforms the benchmark combination.
- ❸ **Economic lesson:** Australian GDP growth is primarily driven by persistence, while the choice of external regressor matters more than adding a second GDP lag or changing the interest-rate specification.

Persistence dominates; terms of trade forecast best out of sample.

A natural next step would be to test longer horizons and formalize the treatment of extraordinary pandemic observations.

Thank you